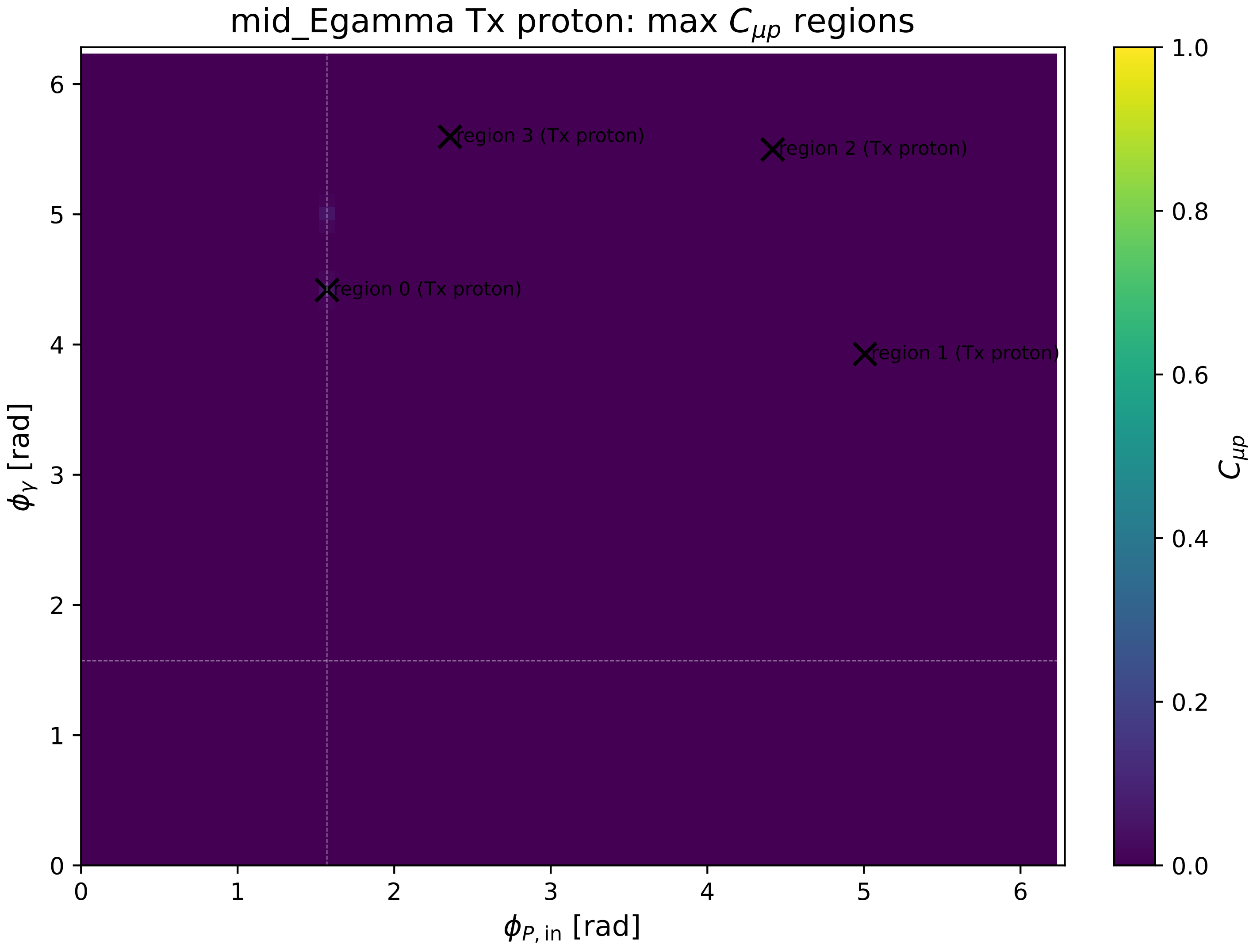
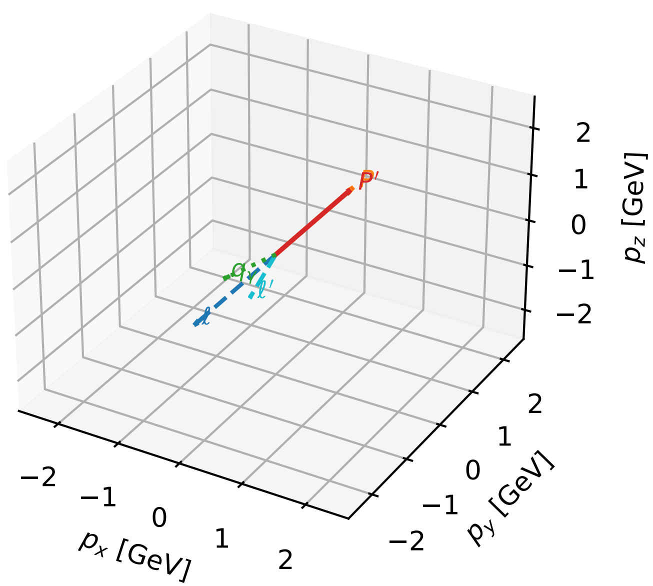

mid_Egamma Tx proton: max $C_{\mu p}$ regions
region 3 (Tx proton)
region 2 (Tx proton)
region 0 (Tx proton)
region 1 (Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx proton characteristic max $C_{\mu p}$ configuration

3D momenta



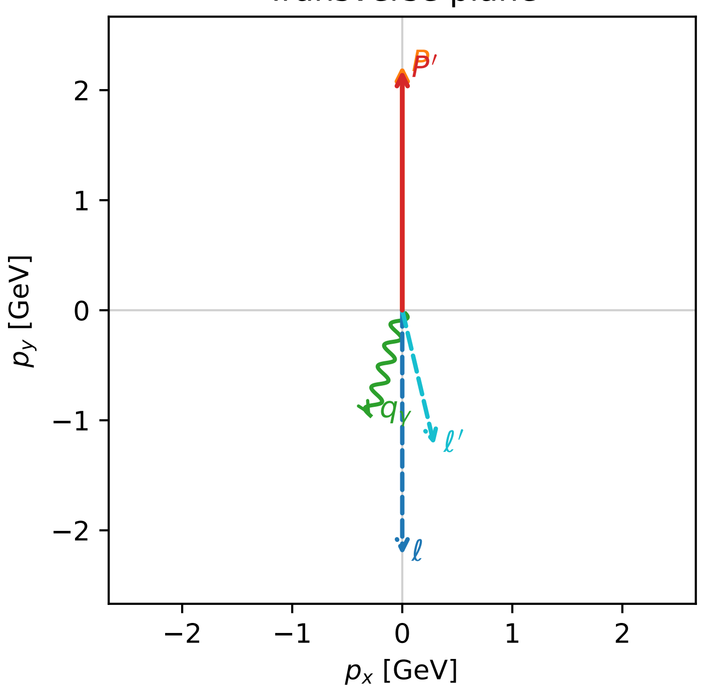

```
c_mu_p_mid_egamma_tx_proton_region_0  C_{mup}=0.0540384
region: mid_Egamma_Tx_proton_region_0  final-pair delta_xy=2.90893 rad
outgoing-state purity: 0.539408
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$

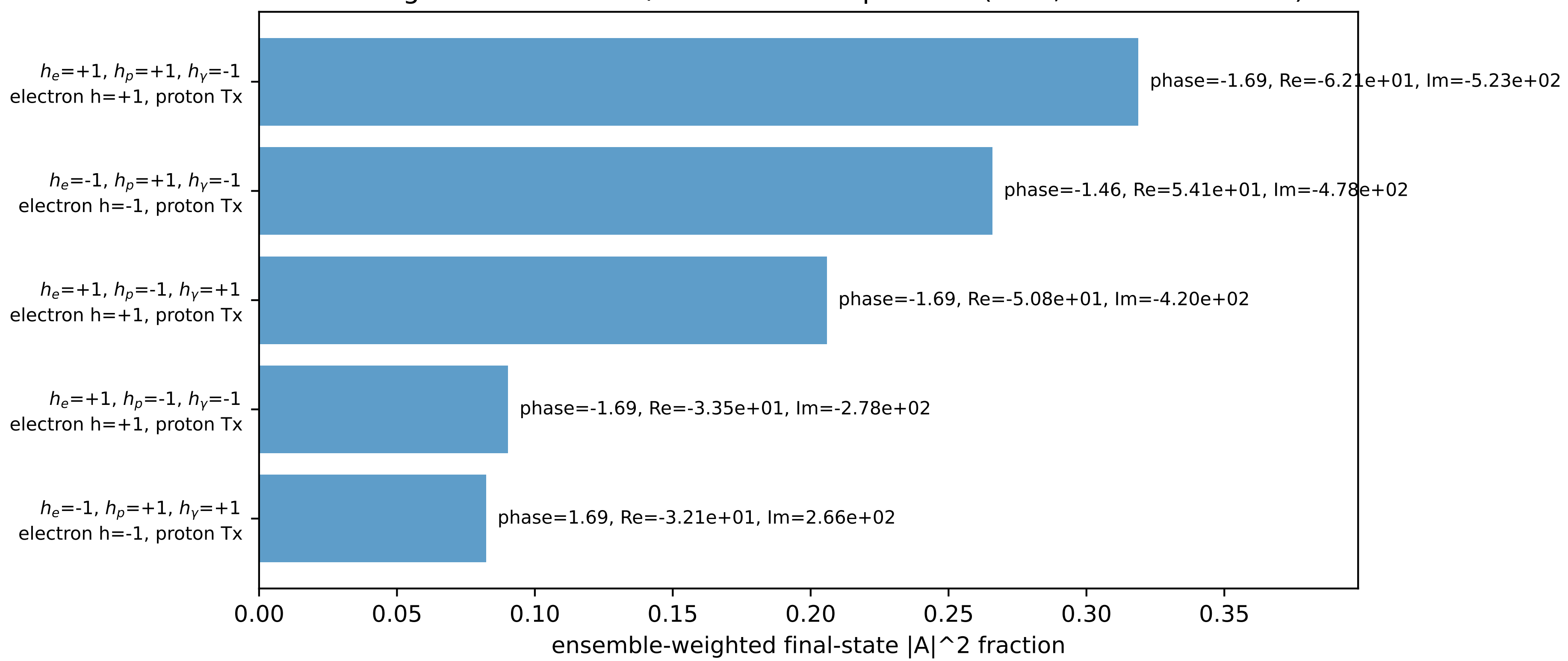
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.18202
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=4.41786
Q^2=0.154517, x_B=0.0170157, t=-0.000259242, W^2=9.80615, y=0.440286
$\theta(\ell', \gamma)$=30.2055 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 1.2634 p_x= 0.2903 p_y= -1.2251 p_z= 0.0000
P' E= 2.3751 p_x= 0.0000 p_y= 2.1820 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.2903 p_y= -0.9569 p_z= 0.0000

Transverse plane

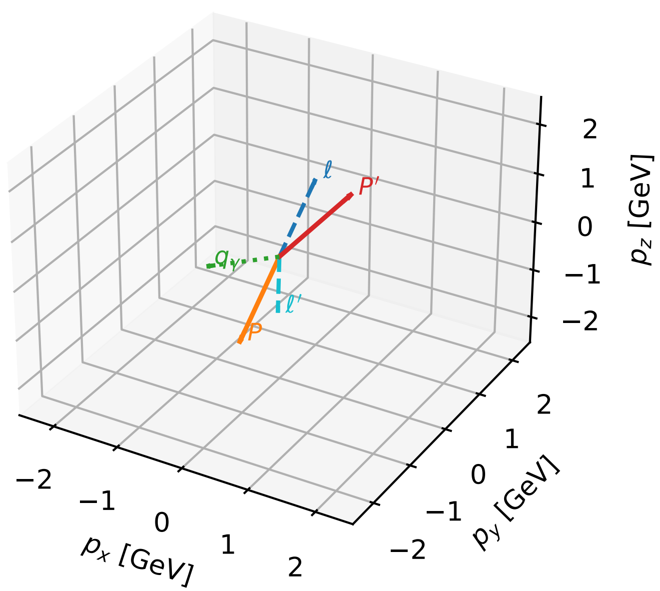


Leading initial-ensemble/final-state components (N=5, retained=96.3%)



Tx proton characteristic max $C_{\mu p}$ configuration

3D momenta



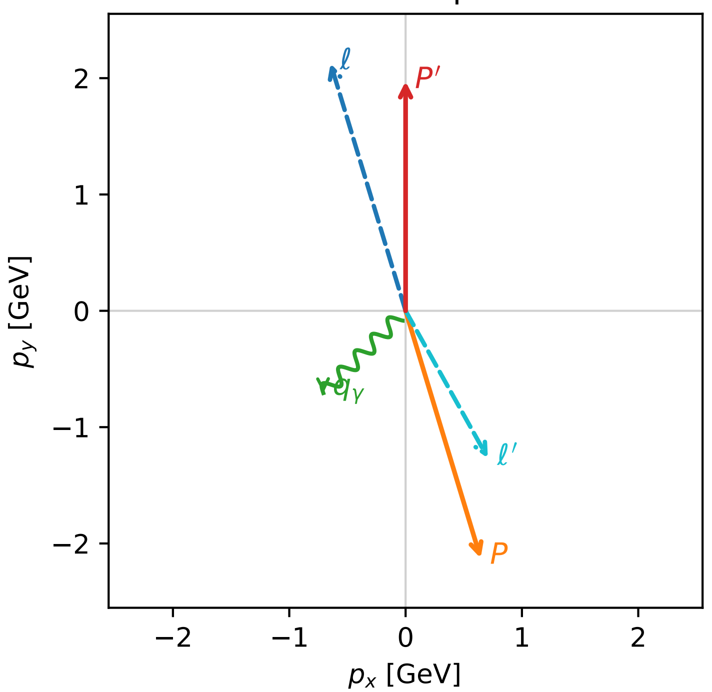

```
c_mu_p_mid_egamma_tx_proton_region_1  C_{mup}=0.00132341
region: mid_Egamma_Tx_proton_region_1  final-pair delta_xy=2.63219 rad
outgoing-state purity: 0.501761
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$

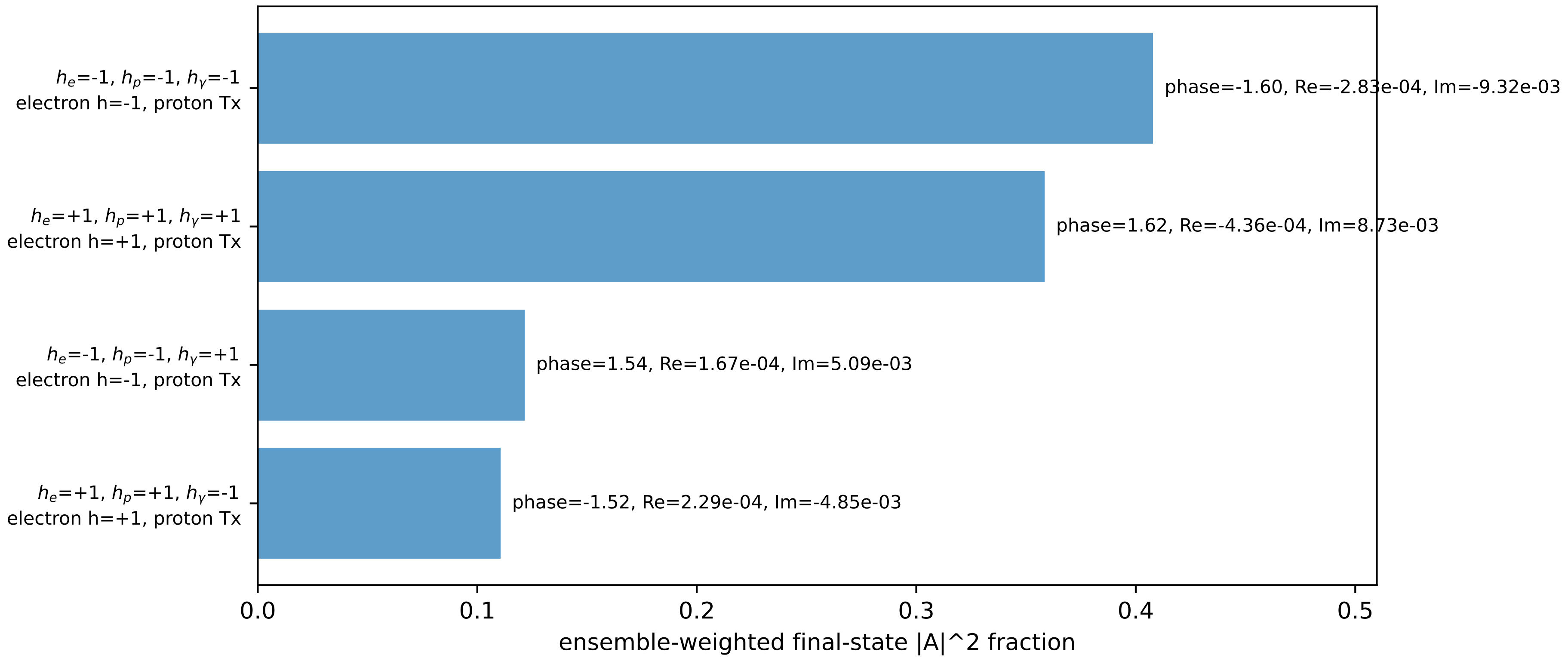
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.97303
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.86532, ϕ_P=5.00691
E_γ=1, ϕ_γ=3.92699
Q^2=12.748, x_B=0.64036, t=-17.1778, W^2=8.03943, y=0.965226
$\theta(\ell', \gamma)$=74.1864 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.6453 p_y= 2.1274 p_z= -0.0000
P E= 2.4129 p_x= 0.6453 p_y= -2.1274 p_z= 0.0000
ℓ' E= 1.4539 p_x= 0.7071 p_y= -1.2659 p_z= 0.0000
P' E= 2.1847 p_x= 0.0000 p_y= 1.9730 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7071 p_y= -0.7071 p_z= 0.0000

Transverse plane

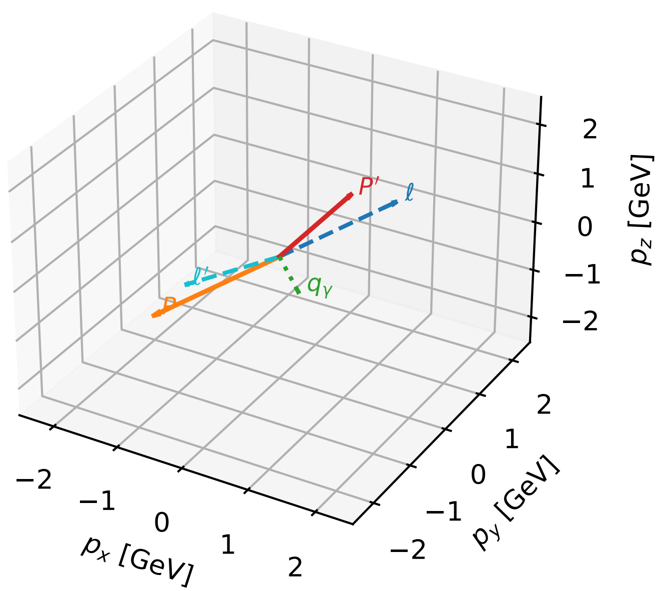


Leading initial-ensemble/final-state components (N=4, retained=99.8%)



Tx proton characteristic max $C_{\mu p}$ configuration

3D momenta



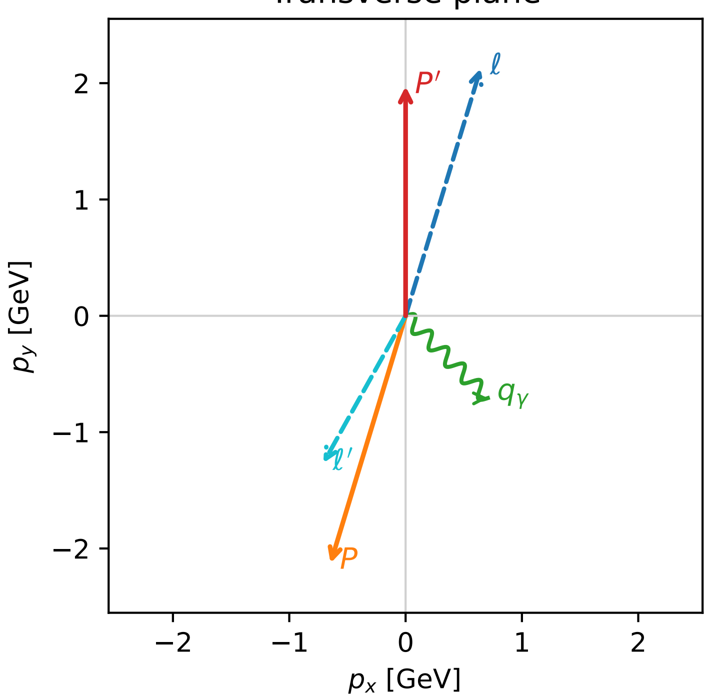

```
c_mu_p_mid_egamma_tx_proton_region_2  C_{mup}=0.00132341
region: mid_Egamma_Tx_proton_region_2  final-pair delta_xy=2.63219 rad
outgoing-state purity: 0.501761
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$

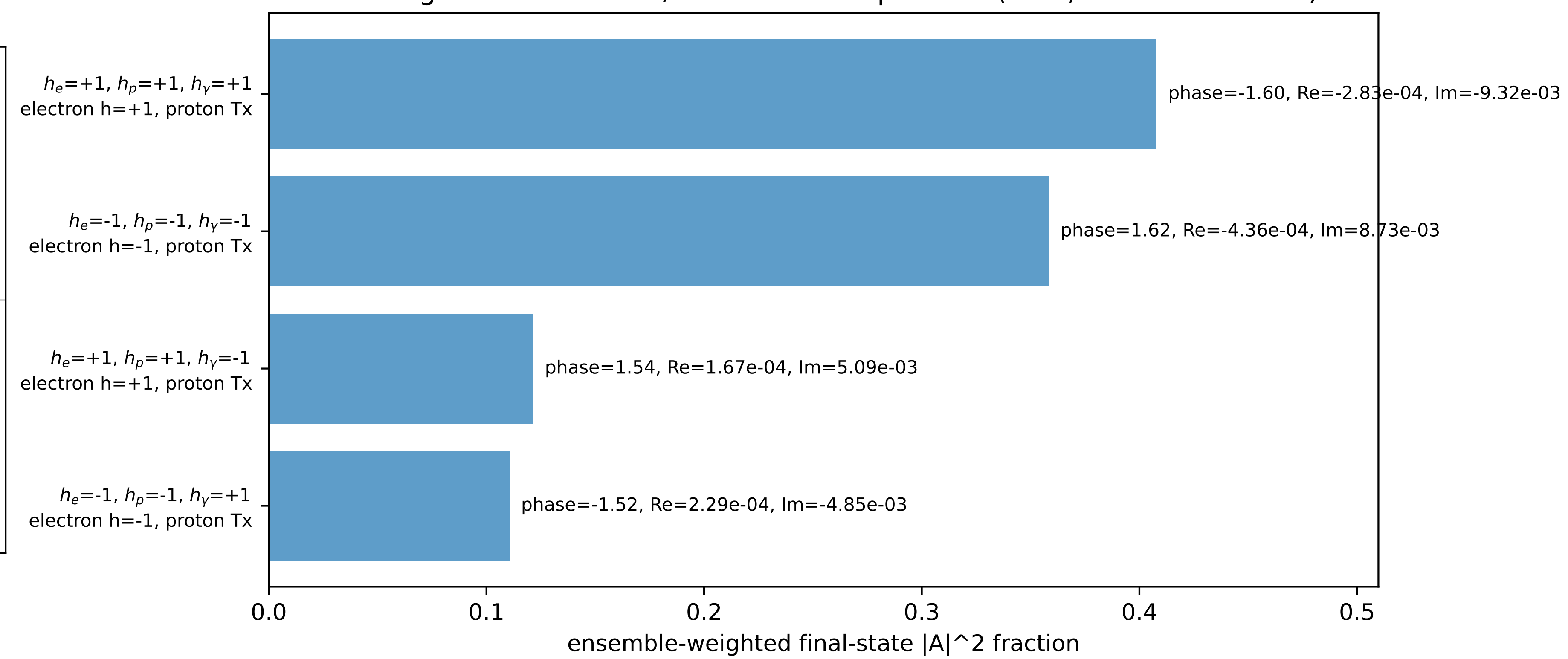
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.97303
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.27627, ϕ_P=4.41786
E_γ=1, ϕ_γ=5.49779
Q^2=12.748, x_B=0.64036, t=-17.1778, W^2=8.03943, y=0.965226
$\theta(\ell', \gamma)$=74.1864 deg

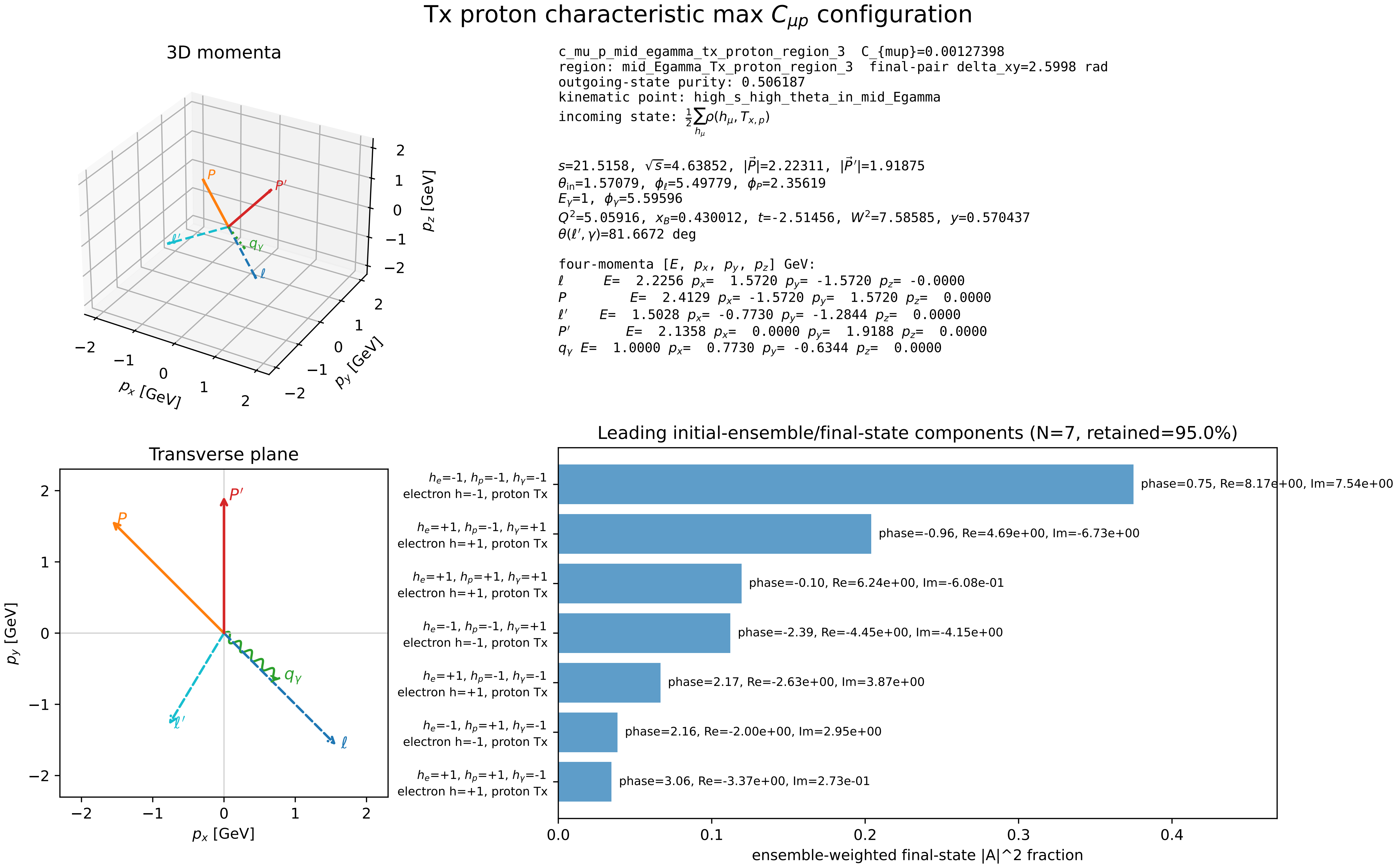
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.6453 p_y= 2.1274 p_z= -0.0000
P E= 2.4129 p_x= -0.6453 p_y= -2.1274 p_z= 0.0000
ℓ' E= 1.4539 p_x= -0.7071 p_y= -1.2659 p_z= 0.0000
P' E= 2.1847 p_x= 0.0000 p_y= 1.9730 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.7071 p_y= -0.7071 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.8%)




Tx proton characteristic max $C_{\mu p}$ configuration
3D momenta
P
P'
ℓ'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
c_mu_p_mid_egamma_tx_proton_region_3 C_{mup}=0.00127398
region: mid_Egamma_Tx_proton_region_3 final-pair delta_xy=2.5998 rad
outgoing-state purity: 0.506187
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.91875
θ_{in}=1.57079, ϕ_ℓ=5.49779, ϕ_P=2.35619
E_γ=1, ϕ_γ=5.59596
Q^2=5.05916, x_B=0.430012, t=-2.51456, W^2=7.58585, y=0.570437
$\theta(\ell', \gamma)$=81.6672 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.5720 p_y= -1.5720 p_z= -0.0000
P E= 2.4129 p_x= -1.5720 p_y= 1.5720 p_z= 0.0000
ℓ' E= 1.5028 p_x= -0.7730 p_y= -1.2844 p_z= 0.0000
P' E= 2.1358 p_x= 0.0000 p_y= 1.9188 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.7730 p_y= -0.6344 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=7, retained=95.0%)
h_e=-1, h_p=-1, h_γ=-1
electron h=-1, proton Tx
phase=0.75, Re=8.17e+00, Im=7.54e+00
h_e=+1, h_p=-1, h_γ=+1
electron h=+1, proton Tx
phase=-0.96, Re=4.69e+00, Im=-6.73e+00
h_e=+1, h_p=+1, h_γ=+1
electron h=+1, proton Tx
phase=-0.10, Re=6.24e+00, Im=-6.08e-01
h_e=-1, h_p=-1, h_γ=+1
electron h=-1, proton Tx
phase=-2.39, Re=-4.45e+00, Im=-4.15e+00
h_e=+1, h_p=-1, h_γ=-1
electron h=+1, proton Tx
phase=2.17, Re=-2.63e+00, Im=3.87e+00
h_e=-1, h_p=+1, h_γ=-1
electron h=-1, proton Tx
phase=2.16, Re=-2.00e+00, Im=2.95e+00
h_e=+1, h_p=+1, h_γ=-1
electron h=+1, proton Tx
phase=3.06, Re=-3.37e+00, Im=2.73e-01
ensemble-weighted final-state |A|^2 fraction